

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Kinds of Data Warehouse Architectures



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- Do you consider important to minimize the quantity of data stored? Why?
- What properties do you consider essential for a data warehouse system?

Can Do Statements:

- I can talk about single layer architecture.
- I can describe second layer architecture.
- I can give information about third layer architecture.

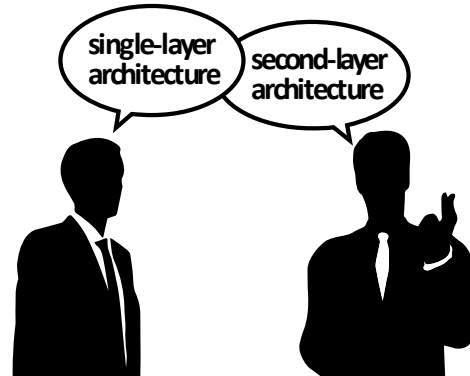




Let's Explore!

Steve is solving Nick's doubts* about data warehousing

*You use "doubts" to refer to questions that someone has about a topic.



Instructions: Read the text below.

Steve: Hi, Nick! You look so stressed, what are you doing?

Nicky: Hi, Steve! I'm studying for tomorrow's exam, but there are some topics that I don't understand. Plus, I need to get a high score if I want to pass this subject because I have a lot of bad grades.

Steve: Which topics don't you get?

Nicky: Basically, it's just data warehouse architecture.

Steve: I studied that topic last semester. So, I can help you if you want.

Nicky: Of course!

Steve: Good! To start, keep in mind some important properties that are crucial for a data warehouse system like separation, scalability, extensibility, security, and administerability. Do you have the function of each one clear?

Nicky: Yes, I know them perfectly. However, I think that my trouble is the three types of layer architecture.

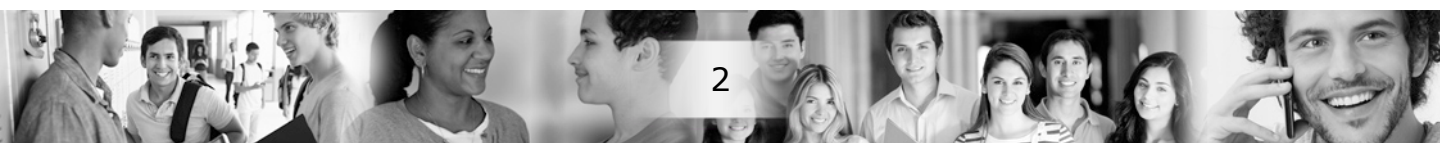
Steve: Don't worry! Firstly, the single-layer architecture permits to minimize the quantity of data the company stores, so it removes data redundancies. In addition, this architecture can use a virtual approach to data warehouses if the volume of data to analyze is huge. However, the failure of this architecture is the separation between analytical and transactional processing. Is it clear?

Nicky: Sure!

Steve: Okay! Secondly, the goal of the two-layer architecture is to separate physical available sources and data warehouses. Also, it brings different benefits to data warehouses such as quality information, specific design solutions, and reliability, among others. Finally, the three-layer architecture allows materializing the operational data left after cleansing and integrating the source data. Moreover, this architecture is very functional, as it creates a common reference data for a whole enterprise, and it separates the problems of source data extraction from problems of data warehouse population.

Nicky: Wow! That was hard, but I understand. I appreciate your help.

Steve: You're welcome, Nicky.





Let's Practice!

Nicky is learning about the single-layer architecture

Instructions: Use the words from the box to correctly complete the sentences.

Word Bank

permits / want / separate / implement / is

1. You need to know first about the single-layer architecture if you _____ to learn about the second and third-layer architectures.
2. If you analyze the single-layer architecture, you learn that it _____ to minimize the quantity of data stored.
3. It removes data redundancies if you _____ a single-layer architecture.
4. The single-layer architecture uses a virtual approach to data warehouses if the volume of data to analyze _____ huge.
5. If you _____ analytical and transactional processes, it is imposible to apply a single-layer architecture.



Let's Practice!

Steve explains more about the second-layer architecture

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. You separate physical available sources and data warehouses if you implement this kind of architecture.
2. If you use this type of architecture, you remove data redundancies.
3. If you use this type of architecture you need to know that it has four subsequent data flow stages.
4. If you apply this kind of architecture, you have different benefits, such as good quality information, specific design solutions, and reliability.
5. If the volume of data to analyze is huge, this architecture uses a virtual approach to data warehouses.
6. If you analyze the single-layer architecture, you see that it permits to minimize the quantity of data stored.

Single-Layer Architecture Features	Second-Layer Architecture Features



Extension Activity

Getting more acquainted with data warehousing!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about architecture and layers in data warehousing.
- Summarize the most important information about the three layers in the warehousing architecture.
- Prepare a presentation with this information using zero conditional and simple conjunctions.
- Make a short presentation to your classmates and instructor.

Self-Assessment

Yes

No

- I research about architecture and layers in data warehousing.
- I identify information about the three layers in warehousing architecture.
- I write about architecture and layers in data warehousing implementing the grammar topics of the level.
- I talk about the most important aspects of architecture and data warehousing.

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Target Words

- data extraction** (noun): process in which data is analyzed to get back relevant information from data sources.
e.g. It separates the problems of source data extraction from problems of data warehouse population.
- data redundancy** (noun): repeated data that you can find in a query result
e.g. Firstly, the single-layer architecture permits to minimize the quantity of data stored, so it removes data redundancies.
- cleanse** (verb): to remove dirt from
e.g. Finally, the three- layer architecture allows to materialize operational data got after cleansing and integrating source data.

- transactional** (adjective): something that is transferable
e.g. However, the failure of this architecture is the separation between analytical and transactional processing.
- administerability** (noun): property of data warehouse system, in which the management of the data warehouse is easier to handle
e.g. Some important properties that are crucial for a data warehouse system like separation, scalability, extensibility, security and administerability.
- trouble** (noun): difficulty to accomplish an activity
e.g. I think that my trouble is the three types of layer architecture.



Learning Points

Zero Conditional

You use **Zero Conditional** when you talk about real or possible situations. When you use the **Zero Conditional**, you have to keep in mind that both clauses (if clause + main clause) are in simple present form. It's important to take into account that you use zero conditional to express a result that always happens.

Use the following rule to make a sentence using zero conditional:

If/When + Simple present + , (comma) + Simple present.

Example:

- **If** the company **has** too many databases, **implement** a data warehouse. Note that you can change the order of the clauses without changing the idea.
- **Implement** a data warehouse **if** the company **has** too many databases.

Examples

- *Plus, I need to get a high score if I want to pass this subject because I have a lot of bad grades.*
- *In addition, this architecture can use a virtual approach to data warehouses if the volume of data to analyze is huge.*

Conjunctions: And, As, Because , But

You use **conjunctions** when you want to connect words, phrases, clauses, or sentences.

Remember that a conjunction is a part of the speech, and it is divided into two groups:

Coordinating Conjunctions can join two different clauses to create a compound sentence. This type of conjunctions gives an equal emphasis to two clauses. In this case, we work on two coordinating conjunctions which are : And – But

You use **And** when you want to link or add one clause to another one.

Example:

*Data warehousing has different architectures, **and** it's really important to differentiate them.*

You use **But** when you want to show contrast between to clauses. Example:





Learning Points

*Data warehousing uses some architectures, **but** it doesn't have data storage on them.*

Note: It's important to use a comma (,) before some coordinating conjunctions when you have two independent clauses.

Subordinating Conjunctions can join clauses, but in this case, one clause becomes dependent upon the other one. Subordinating conjunctions permit to do the necessary transition between one clause to another. In this case, we work on two subordinating conjunctions which are **BECAUSE** – **AS**

You use **Because** and **As** when you want to refer to the **cause** or **reason** for the previous sentence.

Example:

- *Data extraction is really important **because** you can get crucial information from different databases.*
- *The single-layer architecture avoids data redundancies, **as** it minimizes the amount of data stored.*

Examples

- *Moreover, this architecture is very functional, as it creates a common reference data for a whole enterprise.*
- *I'm studying for tomorrow's exam, but there are some topics that I don't understand.*
- *Plus, I need to get a high score if I want to pass this subject because I have a lot of bad grades.*



Learning Points

Data Warehouse Architecture Layer

Data warehouse architecture has commonly two classifications.

It's really important to keep in mind the function of the two classifications of the data warehouse architecture. The first one which gathers three types of layers (single-layer architecture, two layer architecture and three layer architecture) is structure-oriented, and it depends on the amount of layers that the architecture uses. On the other hand, the second classification is determined by the way, in which different layers are used to create enterprise-oriented or department-oriented views of data warehouses.

First Classification	Second Classification
• Single-layer architecture • Two-layer architecture • Three-layer architecture	• Independent data marts architecture • Bus architecture • Hub-and-spoke architecture • Centralized architecture, recommended • Federated architecture